

Specifications for Salmonid Acoustic Telemetry Equipment

Table 1. Receivers and receiver components

Item	Description	Quantity	Date Needed by
1	ATS SR3001 autonomous, tilt sensor, 100-day battery type	17	8/30/2026
2	ATS Replacement Mother Board, SR3001 INCLUDE tilt sensors, Model, 18953, autonomous receiver components	6	10/1/2026
3	ATS Hydrophone Potted Assembly, SR3001, model 17057, autonomous receiver components	6	10/1/2026
4	ATS PVC Battery Connector HM-PN 17860	6	10/1/2026
5	SR3001 replacement battery packs (12 cell), item number 18510	32	8/30/2026
6	Clock battery replacement	7	9/30/2026

Table 2a. Transmitters

ITEM	Tag Type	Quantity
7	Beacon/Reference/417 khz	3
8	JSATS SS400	4,140

Table 2b. Transmitter Specifications and Delivery Schedule

Item	Tag Type	Quantity	Battery Type	PRI	Date Needed by
7	Beacon/Reference/417 khz	3		60	9/30/2026
8a	JSATS SS400	450	TBD	TBD	9/1/2026
8b	JSATS SS400	100	392(2)	5	9/15/2026
8c	JSATS SS400	315	379(2)	TBD	10/1/2026
8d	JSATS SS400	425	BR306	5	11/1/2026
8e	JSATS SS400	630	BR306	7	11/15/2026
8f	JSATS SS400	100	392(2)	5	12/1/2026
8g	JSATS SS400	425	BR306	5	2/1/2027
8h	JSATS SS400	1325	BR306	5	3/1/2027
8i	JSATS SS400	50	BR-306	10	3/1/2027
8j	JSATS SS400	265	392(2)	5	3/1/2027
8k	JSATS SS400	55	379(2)	10	3/1/2027

Ping rate interval (PRI) and battery type to be determined (TBD) by 7/1/26 for items not already specified in the table above.

**Transmitters will be shipped in separate boxes for each row in table 2b.

Schedule

Items should be delivered according to the schedule provided in the need by date in tables 1 and 2b.

Interoperability of equipment: Previous purchases of ATS receivers requires that new shore-based receivers be from the same manufacturer. This specification is important so that if parts of previous equipment breaks (has occurred historically) or is vandalized or stolen (happened in 2020) parts can be exchanged easily in the field without requiring replacing the full system. The receiver array deployment and data systems is more easily maintained through interoperability.

Specifications

JSATS model SS400 ‘injectable’

Must meet requirements ensuring that the vendor is capable of producing transmitters to the specifications and of sufficient quality, functionality, and reliability.

Transmitters must be manufactured in the USA.

Transmitter must have capability to be user configurable in the field using a portable laptop PC connected to a vendor supplied tag controller/activator. Re-programming parameters must include: Tag code value (4 digit hexadecimal number) and PRI (Pulse Rate Interval) between 2 and 60 seconds.

The following values must be able to be displayed and verified (but not necessarily changed) in the field using the vendor supplied portable tag controller/activator in stand-alone configuration: (1) Tag code value (4 digit hexadecimal number), (2) CRC value (cyclic redundancy check), (3) PRI Measured (pulse rate interval, in seconds with 1000th of a second precision) and (4) Estimated tag frequency (in kHz).

Tag shall be activated (started) by using an Advanced Telemetry System’s Pinger Dish, if tags cannot be started with the ATS Pinger dish, then three (3) tag activators that will work must be added to the quote. Tag activators must have a serial or USB connection to a PC, so that data from tags can be recorded to a file on the PC.

The tag’s PRI may be changed by user at time of activation and the warranty life of that tag will be based on the warranty life for PRI stated in proposal.

Warranty tag life shall be 50% of advertised expected life for the set PRI.

Tags started 1 to 180 days after delivery date shall be expected to run for at least 100% of stated warranty life.

Tags started 180 to 365 days after delivery date shall be expected to run for at least 90% of stated warranty life.

Tags started more than 365 days after delivery date shall be expected to run for at least 75% of stated warranty life.

Warranty tag life for each of these pulse rate intervals; 3, 4.2, 5, 7, 10, and 60 seconds, shall be stated in quote.

5% of transmitters for total order will be randomly selected and monitored to determine actual run time. Tags that do not run past warranty life will be considered as failed. If percent failed is greater than 6% of

tags from the 5% subsample then % failed shall be scaled up to total number of tags purchased and warranty claim shall be based on this number.

Vendor shall provide an MS Excel format electronic file of specifications for each tag including tag model, battery model, hexadecimal ID, nominal PRI, measured tag weight in air, tag dimensions (length, width, and height), power output, warranty life at the specified PRI, and manufacture date.

JSATS compatible – binary phase shift keyed code pulse operating on frequency of 416.7 kHz +/- 0.5%. Possible code set of 65,000 individual IDs

Power output of at least 154 dB re: 1uPa @ 1 meter

Labeled with 4 place alphanumeric code ID

Specific code IDs selected by user and provided to manufacture at time of award.

Coating of Parylene C, 25 micron thickness

Maximum weight in air of 0.22 grams

Maximum size of 15 mm long x 3.4 mm diameter dimensions

Autonomous Receivers

Depth ratings: 70 m, minimum

Operating Temperature: 0 – 50 °C

Data storage capacity: Two gigabytes of removable memory

Service life: 45 days

Size: 15 cm diameter, 76 cm long (6 x 30 in.

Weight: 4.5 kg (10 lbs)

Buoyancy: Positively buoyant

Modes of Operation: Receiver detects, decodes, and stores to memory all valid tag codes. Additionally, the receiver records environmental information including ambient external temperature (°C), and deviation from vertical (degrees).

Diagnostic Information: Receiver records at user defined intervals: battery voltage, and number of beacon tags detected since last diagnostic check.

Optional Sensors: Ambient external temperature (°C), and position in digress relative to the vertical hydrophone position (tilt)

Communications: Direct/cabled communications: Direct connection to receiver via USB allows user to set up deployment site identification, clock synchronization, access to stored data, and links to diagnostic sensors.

PC User interface: Included software provides a graphical user interface to retrieve data and perform setup functions

Warranty: One year parts and labor on materials and workmanship